



Effects of reduced metabolizable protein diets supplemented with rumen-protected His under adequate Met and Lys supply on production, nutrient digestibility, N utilization, and manure ammonia emissions in lactating cows

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ABSTRACT

Balancing for limiting AA may prevent reduced production when deficient MP is provided. The objective of the study was to determine the effects of rumen-protected (RP) His supplementation at different levels of MP when Lys and Met supplies meet the requirements. Sixty lactating cows (mean $\pm$ SD; parity, 2.47 ± 1.346 ; DIM, 112 ± 27 ; milk yield, 50 ± 8 kg/d) were used in a randomized complete block design and allocated to one of the following diets: an MP-adequate diet meeting the MP, Met, Lys, and His requirements (CP, 17.0%); an MP-moderately deficient diet (92.6% of the requirement) meeting the requirement of Met and Lys (MDMP; CP, 15.4%); the MDMP diet supplemented with RP-His to meet the His requirement; an MP-deficient diet (86.7% of requirement) meeting the requirement of Met and Lys (DMP; CP, 14.5%); and the DMP diet supplemented with RP-His to meet the requirement of His. The experiment lasted for 6 weeks of data collection after 2 wk of covariate. During the last week, feces and urine collected from 8 blocks were used to reconstitute manure based on the individual daily excretion of feces and urine. Manures were incubated for 12 d using a steady-state air-flux chamber system. Milk yield, milk fat and protein, DMI, and BW were not affected by treatments. Lowering MP supply linearly decreased total-tract N digestibility and urinary N excretion but increased N use efficiency for milk protein (MNE; 31.4% to 38.3% of N intake). Lowering MP supply also linearly decreased plasma urea-N (11.2 to 8.12 mg/dL) and tended to decrease plasma His (57.6 to 48.0 μ mol/L). Supplementation of RP-His decreased MNE because it did not increase milk N excretion. Blood hemoglobin (8.21 to 8.40 g/dL) and plasma His (50.4 to 52.2 μ mol/L) and carnosine (14.4 to 14.9 μ mol/L) numerically increased by RP-His, but the differ-

ences were not significant. Lowering MP supply tended to linearly increase the ratio of feces to urine and linearly decreased daily and cumulative NH₃ emissions from manure. However, RP-His supplementation did not affect daily and cumulative NH₃ emissions. In conclusion, the diets supplying deficient MP but adequate Met and Lys did not negatively affect milk yield and protein yield. Supplementing the MP-deficient diets with RP-His did not increase milk and milk protein yields. It is likely that additional His was provided from the endogenous His pool during the 6-wk feeding period, resulting in lack of His deficiency. Further studies on the sources of endogenous His and their contribution to milk protein synthesis under varying degrees of His deficiency.

Key words: amino acids, histidine, low-protein diet, ammonia

INTRODUCTION

Achieving sustainable dairy production requires both improving profitability and reducing the environmental effects of dairy farming (VandeHaar and St-Pierre, 2006). Dietary protein source is one of the expensive ingredients in dairy rations, and inefficient utilization of dietary protein results in less profitability from the dairy herd (Ishler and Beck, 2023) and potential contribution to various environmental pollutions due to N excretion in manure. For example, in the United States, the agriculture sector accounts for 9.4% of total GHG emissions, and N₂O contributes the largest portion (52%; 308 million tonnes of CO₂ equivalents) of the GHG produced in the agriculture sector in 2022 (EPA, 2024). The N₂O emission from dairy manure was estimated to be 6.2 million tonnes of CO₂ equivalents of N₂O, which accounts for 36.5% of total N₂O emitted from livestock manure in the United States. In addition, livestock manure is the largest source of anthropogenic ammonia in the United States (USEPA, 2004). Ammonia is a pollutant that can contribute to particulate matter formation, water and soil eutrophication, and indirect N₂O emission (Hristov et al., 2011). There-

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

fore, improving dietary N use efficiency for milk protein synthesis (**MNE**; milk N $\div$ N intake) has been one of the key goals for sustainable dairy production.

To increase the MNE and reduce manure nitrogen emissions, reducing dietary CP concentration is probably the most effective strategy (Huhtanen and Hristov, 2009; Sajeev et al., 2018). However, it has always been debatable how much dietary CP can be reduced without compromising animal performance. This is because reduced CP in a diet below the protein requirement has often resulted in decreases in milk and protein yields in previous studies (Cyriac et al., 2008; Lee et al., 2011). To prevent the performance losses, rumen-protected (**RP**) Met and Lys have been supplemented to low-protein diets in research. The results were somewhat successful (i.e., no performance losses; Broderick et al., 2008; Robinson et al., 2010; Lee et al., 2012c). However, there are studies that showed decreases in milk or milk protein yields when lactating cows were fed reduced-CP diets supplemented with RP-Lys and Met (Lee et al., 2012b; Giallongo et al., 2016). The inconsistent responses observed across studies might be due to varying degrees of potential His deficiency. When a diet deficient in MP is fed, the contribution of microbial protein (**MCP**) to total metabolizable AA increases (Lee et al., 2012b,c). Then, His can be another limiting AA because of relatively low concentration of His in MCP (Sok et al., 2017). Therefore, feeding a reduced-CP diet supplemented with RP-His in addition to RP-Met and RP-Lys has been studied, and increases in milk and milk protein yield were observed (Lee et al., 2012c; Giallongo et al., 2016; Räisänen et al., 2023). However, studies with no effect of feeding RP-His on performance are available as well (Räisänen et al., 2021b, 2022; Van den Bossche et al., 2023). The degree of the deficiency of MP varies among studies (85% to 97% of the MP required according to NRC, 2001; Lee et al., 2012c; Giallongo et al., 2016). A meta-analysis by Räisänen et al. (2023) reported that the response in milk protein concentration and yield to RP-His supplementation depends on the dietary MP supply.

The objective of this study was to determine the effects of different levels of MP deficiency and RP-His supplementation on production when metabolizable Lys (**mLys**) and metabolizable Met (**mMet**) supplies meet the requirements. We hypothesized that the MP deficiency would increase MNE and decrease urinary N excretion and NH_3 emissions from manure. However, the supply of deficient MP will decrease milk protein yield although mMet and mLys supplies meet the requirements, and the decreased milk protein yield by MP deficiency will be alleviated by RP-His supplementation. Also, it was hypothesized that ammonia emissions from manure will not be affected by RP-His supplementation, whereas it will decrease linearly by decreasing MP.

MATERIALS AND METHODS

Animals, Design, and Diet

All experimental procedures and animal care were approved by the IACUC of The Ohio State University (2023A00000021). A total of 60 early- and mid-lactation cows (average $\pm$ SD; 112 ± 27 DIM; 50 ± 8 kg/d milk yield; 665 ± 84.5 kg BW; 27.5 ± 3.50 kg DMI) were used in a randomized complete block design (2/3: multiparous; 1/3: primiparous). Cows were blocked by parity, DIM, and milk yield and allocated to 12 blocks (5 cows each). In each block, cows were randomly assigned to one of the following diets: an MP-adequate diet supplemented with RP-Met (Smartamine-M, Adisseo Inc.; 75% Met with an assumed bioavailability of 80%) that meets the MP, mMet, mLys, and metabolizable His (**mHis**) requirements (**AdMP**) according to NASEM (2021); an MP-moderately deficient diet supplemented with RP-Met and Lys (AjiPro-L, Ajinomoto Co. Inc.; 40% Lys with an assumed bioavailability of 60%) that meets 93% of MP, 105% of mMet and mLys, and 91% of mHis required (**MDMP**); MDMP diet with 45 g/d of RP-His (prototype, Ajinomoto Co. Inc.; 42% His with an assumed bioavailability of 50%) to meet the His requirement (**MDMPH**); an MP-deficient diet supplemented with RP-Met and Lys that meets 85% of MP, 105% of Met and Lys, and 81% of His required (**DMP**; CP, 14.5%); and MDP diet with 79 g/d of RP-His to meet the requirement of His (**DMPH**). To lower the supply of MP, the proportions of the major protein ingredients (soybean meal, heat-treated soybean meal, blood meal) gradually decreased by gradually increasing the proportions of corn grain and soybean hulls. The major ingredients (corn silage, alfalfa silage, corn grain, soybean meal, heat-treated soybean meal, blood meal, and soybean hulls) were analyzed for AA composition at the Agricultural Experiment Station Chemical Laboratories, University of Missouri-Columbia, and the results were used to estimate metabolizable AA supply and requirements using NASEM (2021) to determine the inclusion rate of RP-Met, -Lys, and -His. The diet formulation and chemical composition are shown in Tables 1 and 2. The experiment was conducted for 8 wk. During the first 2 wk, all cows were fed the AdMP diet, and the performance data in wk 2 were used as a covariate. After the covariate period, cows received the experimental diets for 6 wk. Diets were prepared every morning and fed to cows once daily at 0700 h. All cows were housed in individual tiestalls and fed the diets for ad libitum intake (the target refusal rate of 5%), with free access to water. Cows were milked twice daily at 0400 and 1600 h. Rumen-protected Lys and Met were included in the diets, and RP-His was top-dressed immediately after the diets were delivered (MDMPH and DMPH).

Table 1. Diet ingredients and composition

Item	Diet ¹				
	AdMP	MDMP	MDMPH	DMP	DMPH
Ingredient (DM%)					
Corn silage	40.0	40.0	40.0	40.0	40.0
Alfalfa silage	14.5	14.5	14.5	14.5	14.5
Corn grain dry, fine grind	13.0	13.8	13.8	14.1	14.2
Soybean meal	7.94	7.54	7.54	7.14	7.14
Heat-treated soybean meal ²	7.80	4.60	4.60	2.60	2.60
Blood meal ³	0.80	0.60	0.60	0.40	0.40
Soybean hulls	11.1	13.7	13.7	15.7	15.7
Fat supplement ⁴	1.72	1.72	1.72	1.72	1.72
Vitamin and mineral mix	3.07	3.07	3.07	3.07	3.07
RP-Met ⁵	0.10	0.13	0.13	0.15	0.15
RP-Lys ⁶	0.00	0.36	0.34	0.62	0.59
RP-His ⁷	0.00	0.00	0.18	0.00	0.31
Chemical composition (% DM)					
OM	93.8	93.6	93.8	93.9	93.9
NDF	28.4	28.9	28.6	30.5	30.4
CP	17.0	15.4	15.8	14.5	14.7
RDP ⁸	10.2	9.7	9.7	9.4	9.4
RUP ⁸	6.9	5.8	5.8	5.1	5.1
Starch	24.9	25.4	26.3	25.1	24.7
Fatty acids	4.30	4.45	4.47	4.48	4.43
Ca	0.65	0.69	0.67	0.65	0.69
P	0.34	0.31	0.33	0.30	0.28

¹AdMP, an MP-adequate diet that meets the MP, Met, Lys, and His requirements (NASEM, 2021); MDMP, an MP-moderately deficient diet (92.6% of the requirement) meeting the requirement of Met and Lys; MDMPH, the MDMP diet with RP-His to meet the His requirement; DMP, an MP-deficient diet (86.7% of requirement) meeting the requirement of Met and Lys; DMPH, the DMP diet with RP-His to meet the requirement of His.

²AminoPLUS, Consolidated Nutrition LC.

³AAdvantage, Perdue AgriBusiness.

⁴EnergyBooster-100, Milk Specialties Co.

⁵Smartamine-M, Adisseo Inc.

⁶AjiPro-L, Ajinomoto Health & Nutrition North America Inc.

⁷Prototype, Ajinomoto Health & Nutrition North America Inc.

⁸Estimated using NASEM (2021).

Sampling and Analysis

Samples of concentrates, corn silage, and alfalfa silage were collected weekly during the entire experiment. Weekly samples of forages were dried at 55°C for 72 h to determine DM. Dried samples from the experimental period were composited for the following chemical analyses. The dried samples were ground through a 1-mm sieve (ZM200, Retsch, Germany) and analyzed for OM (method 942.05; AOAC International, 2000), NDF with heat-stable α -amylase and sodium sulfite (Ankom 200 Fiber Analyzer; Ankom Technology Co., Macedon, NY), fatty acids (Jenkins, 2010), total N using an elemental analyzer (Flash 2000, Thermo Fisher Scientific, Waltham, MA), starch (Hall, 2009), and minerals (Rock River Laboratory, Watertown, WI) using inductively coupled plasma emission spectroscopy. Refusal samples were collected weekly for 2 consecutive days except for the adaptation period. The refusal samples were composited by week and dried at 100°C for 72 h to calculate DMI.

Milk yields were recorded daily for individual cows during the entire experiment. Milk samples were collected from individual cows on 2 consecutive days weekly and analyzed for milk fat, true protein, lactose, and MUN by DHI Cooperative Inc. (Columbus, OH) using infrared analysis (B2000, Bentley Instruments, Chaska, MN). Additional milk samples on wk 8 were collected and analyzed for total N using a combustion method (Flash 2000, Thermo Fisher Scientific, MA).

Blood was collected 6 h after feeding on wk 2 and 8 from the tail vein using lithium-heparin vacutainers (45584, Vacuette, Greiner Bio-One GmbH). The samples were immediately analyzed for Na, K, Cl, iCa, glucose, BUN, creatinine, hematocrit (**Hct**), total CO₂, anion gap, and hemoglobin (**Hb**) using iSTAT Alinity v with the cartridge of Chem8+ analyzer (Abbott Point of Care). The remaining blood was centrifuged at 2,500 × g for 15 min at 4°C to collect plasma. The plasma samples were determined for albumin, alkaline phosphatase, aspartate aminotransferase, Ca²⁺, creatine kinase, gamma-glutamyl transferase, globu-

Table 2. Net energy for lactation, MP, and AA balances estimated from NASEM (2021)

Item	Diet ¹				
	AdMP	MDMP	MDMPH	DMP	DMPH
NEL, Mcal/d					
Supply	49.3	48.5	48.5	48.2	48.6
Requirement	45.4	45.7	44.0	44.1	44.9
Balance	3.90	2.82	4.45	4.09	3.72
MP, g/d					
Supply	2,943	2,620	2,629	2,410	2,441
Recommended	2,867	2,843	2,826	2,803	2,792
Balance	76	−223	−197	−393	−351
mHis, g/d					
Supply	77	66	75	58	74
Recommended	75	75	74	74	73
Balance	2.0	−9.0	1.0	−16	1.0
mMet, g/d					
Supply	74	72	72	72	72
Recommended	71	71	70	69	69
Balance	2.5	0.8	1.8	3.0	3.2
mLys, g/d					
Total supply	226	224	224	223	225
Recommended	221	220	219	217	215
Balance	5.0	3.7	4.7	6.0	9.6

¹AdMP, an MP-adequate diet that meets the MP, Met, Lys, and His requirements (NASEM, 2021); MDMP, an MP-moderately deficient diet (92.6% of the requirement) meeting the requirement of Met and Lys; MDMPH, the MDMP diet with RP-His to meet the His requirement; DMP, an MP-deficient diet (86.7% of requirement) meeting the requirement of Met and Lys; DMPH, the DMP diet with RP-His to meet the requirement of His.

lin, Mg, inorganic phosphorus, total protein, and BUN using VetScan with the Large Animal Profile rotor (Abaxis Inc.). Plasma AA concentrations were determined using ultra-performance liquid chromatography mass spectrometry (Waters Corp.) with the AccQ-Tag AA derivatization according to the manufacturer's protocol (Waters Corp.). An internal standard of DL-Norleucine was used to correct the AA concentrations in samples.

Spot urine and feces were collected on wk 8. A total of 6 spot samples of feces and urine per cow were collected over 3 d to represent every 4 h sampling in a 24 h cycle (i.e., 1000 and 2200 h on d 1; 0600 and 1800 h on d 2; 0200 and 1400 h on d 3) according to the procedures by Morris et al. (2018) and Lee et al. (2019). Spot urine samples were collected by massaging the escutcheon, and spot fecal samples were collected from the rectum. The collected samples were composited by cow and split into 2 parts: one part for chemical analyses and another part for manure incubation.

Urine samples for chemical analyses were acidified using 2 M H₂SO₄, and frozen until further analysis. The samples for manure incubation were kept at −20°C without acidification for manure incubation later. The acidified urine samples were assayed for creatinine (kit 500701, Cayman Chemical), allantoin (Chen et al., 1992), uric acid (Stanbio Uric Acid Kit 1045; Stanbio Laboratory Inc.), urea-N (Kit UR107; Randox Laboratories Ltd.), and total N using an elemental analyzer (Flash 2000, Thermo Fisher Scientific). Creatinine was used to

estimate the daily urine outputs of individual cows using 23 of the coefficients for the urinary creatinine excretion rate (mg/kg BW·d^{−1}; Park et al., 2025).

The fecal samples for individual analysis were dried at 55°C for 3 d to determine DM. Dried samples were ground through a 1-mm sieve and were analyzed for DM, OM, NDF, and N as described earlier for feeds. The chemical composition of feces was used to calculate the total-tract digestibility of DM, OM, NDF, and N. Indigestible NDF (**iNDF**) of the concentrates, corn silage, alfalfa silage, and feces were analyzed using 2 cannulated cows fed the herd lactation diet (forage: concentrate = 49: 51; 16.7% CP, 30.8% NDF, and 25.6% starch, DM basis). Dried and ground samples were prepared in filter bags (F57; Ankom Technology Co.) and incubated in the rumen for 12 d (triplicate samples per cow). After incubation, the bags were washed and assayed for NDF as for feeds. The residuals were considered iNDF and its concentration was used to estimate fecal DM outputs of individual cows (Huhtanen et al., 1994).

Eight blocks of cows were randomly selected for the manure incubation. Feces and urine samples collected for manure incubation were mixed to reconstitute manures according to the daily outputs of urine and feces estimated for individual cows. The manures were incubated to determine the CH₄, CO₂, and NH₃ emissions from the manure using the manure incubation system as described in Lee et al. (2020). The incubation lasted for 12 d, and gas samples from individual manure chambers were col-

Table 3. Effects of MP supply and rumen-protected His supplementation under adequate Met and Lys supplies on DMI, milk production, milk composition, feed efficiency, and BW change in lactating cows

Item ³	Diet ¹					SEM	P-value ²			
	AdMP	MDMP	MDMPH	DMP	DMPH		Overall	L	Q	His
DMI, kg/d	27.7	27.4	27.4	27.3	27.5	0.36	0.96	0.52	0.82	0.83
Milk yield, kg/d	45.6	44.7	44.9	45.2	44.3	0.64	0.53	0.57	0.35	0.49
Milk yield/DMI, kg/kg	1.65	1.65	1.64	1.67	1.61	0.026	0.67	0.68	0.69	0.25
ECM, kg/d	45.8	47.1	44.9	44.7	45.9	1.08	0.54	0.47	0.17	0.68
ECM/DMI, kg/kg	1.67	1.71	1.65	1.67	1.67	0.037	0.65	0.92	0.24	0.31
Fat, kg/d	1.52	1.65	1.45	1.49	1.54	0.08	0.40	0.80	0.12	0.33
Protein, kg/d	1.49	1.47	1.47	1.45	1.44	0.024	0.53	0.20	0.92	0.85
Lactose, kg/d	2.23	2.18	2.19	2.22	2.17	0.037	0.66	0.92	0.26	0.53
Fat, %	3.45	3.79	3.41	3.34	3.62	0.218	0.49	0.68	0.11	0.80
Protein, %	3.29	3.34	3.30	3.24	3.27	0.041	0.50	0.46	0.12	0.85
Lactose, %	4.89 ^{ab}	4.87 ^b	4.88 ^{ab}	4.93 ^a	4.90 ^{ab}	0.019	0.20	0.11	0.07	0.70
MUN, mg/dL	13.1 ^a	10.9 ^b	10.6 ^b	10.0 ^b	9.84 ^b	0.41	<0.01	<0.01	0.25	0.57
Initial BW, kg	684	666	646	655	674	10.9	0.35	0.15	0.84	0.96
End BW, kg	691	684	657	669	678	10.6	0.54	0.31	0.84	0.56
BW change, kg/d	0.161	0.416	0.246	0.327	0.109	0.0584	0.47	0.37	0.29	0.14

^{a,b}Means in a row with differing superscripts differ ($P < 0.05$).

¹AdMP, an MP-adequate diet that meets the MP, mMet, mLys, and mHis requirements; MDMP, an MP-moderately deficient diet (92.6% of the requirement) meeting the requirement of mMet and mLys; MDMPH, the MDMP diet supplemented with RP-His to meet the mHis requirement; DMP, an MP-deficient diet (86.7% of requirement) meeting the requirement of mMet and mLys; DMPH, the DMP diet supplemented with RP-His to meet the requirement of mHis.

²Overall, main effect of the diets; main effect of week was significant in all parameters ($P < 0.05$); L, linear effect of the MP supply (AdMP vs. MDMP vs. DMP); Q, quadratic effect of the MP supply (AdMP vs. MDMP vs. DMP); His, effect of RP-His supplementation (MDMP, DMP vs. MDMPH, DMPH).

³An interaction between treatment and week was observed for lactose concentration ($P = 0.04$) and MUN ($P < 0.01$).

lected into a Tedlar sampling bag (22053, RESTEK Co.) for 30 s every 30 min for 24 h on d 1, 2, 3, 4, 6, 8, and 12. Gas samples collected were analyzed using a photo-acoustic gas analyzer (Gasera one, Gasera Ltd.). Daily gas production from manure was calculated according to the following equation with an example of NH_3 :

$$\begin{aligned} \text{NH}_3 \text{ g/cow} \cdot \text{d}^{-1} &= [\text{NH}_3 \text{ concentration (g/L)} \\ &\quad - \text{blank NH}_3 \text{ concentration (g/L)}] \\ &\quad \times \text{air flow (2.0 L/min)} \times 1,440 \times \text{density} \\ &\quad (0.769 \text{ g/L of NH}_3 \text{ at standard temperature and pressure)} \\ &\quad \times \text{estimated daily manure output.} \end{aligned}$$

The area under the curve from the daily gas emissions was calculated for individual cows to obtain cumulative gas emissions for 12 d.

Statistical Analysis

Data were analyzed using the Mixed Procedure of SAS 9.4 software (SAS Institute, Cary). Data of DMI, milk yield and composition, and daily manure gas emission were analyzed using the following model:

$$Y_{ijk} = \mu + (\text{Cov}) + D_j + T_k + (\text{DT})_{jk} + b_i + e_{ijk},$$

where Y_{ijk} = dependent variable; μ = overall mean; Cov = measures during the covariate period (only applied for DMI and milk production); D_j = fixed effect of dietary treatment ($j = 1$ to 5); T_k = fixed effect of time ($k = 1$ to 6 for production data; $k = 1$ to 12 for daily manure gas emission data); $(\text{DT})_{jk}$ = the interaction of between diet and time; b_i = random effect of block ($i = 1$ to 12); and e_{ijk} = residual error. Time was used for the repeated measure, and the covariance structure with the lowest Akaike information criterion value among UN, CS, CSH, AR(1), ARH(1), and ANTE(1) was used. The Kenward-Roger method was applied to calculate the denominator df. The other data were analyzed using the model above, excluding the fixed effect of time and the interaction between treatment and time. Preplanned contrasts were applied to determine the linear and quadratic effects of MP supply (AdMP vs. MDMP vs. DMP), and effects of RP-His supplementation (MDMP, DMP vs. MDMPH, DMPH). All data were presented as LSM. Significance was declared at $P < 0.05$, and tendency at $0.05 \leq P < 0.10$.

RESULTS

Dry matter intake, milk yield, milk component concentrations and yields, and BW were not affected by treatments during the experiment (Table 3). An interaction

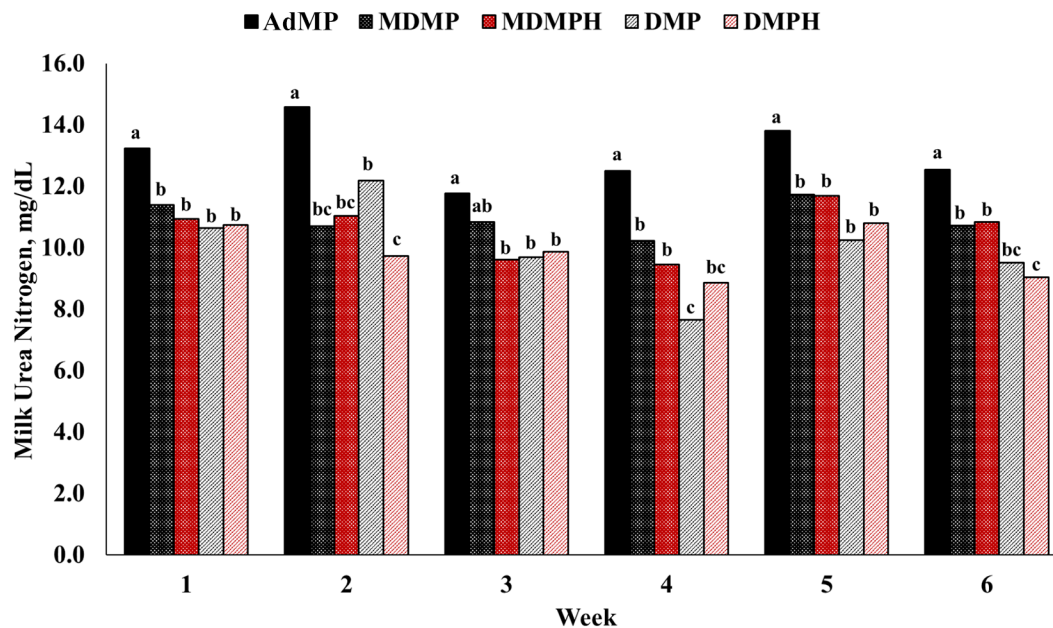


Figure 1. Effects of MP supply and rumen-protected His supplementation under adequate Met and Lys supplies on MUN in lactating cows. Main effect of diets, week, and interaction between diet and week were significant ($P < 0.01$). Linear effect of decreasing MP supply was observed (AdMP vs. MDMP vs. DMP; $P < 0.01$). Bars without a common letter within a week are significantly different ($P < 0.05$). AdMP, an MP-adequate diet that meets the MP, Met, Lys, and His requirements (NASEM, 2021); MDMP, an MP-moderately deficient diet (92.6% of the requirement) meeting the requirement of Met and Lys; MDMPH, the MDMP diet with RP-His to meet the His requirement; DMP, an MP-deficient diet (86.7% of requirement) meeting the requirement of Met and Lys; DMPH, the DMP diet with RP-His to meet the requirement of His.

between treatment and wk was observed for MUN ($P < 0.01$; Figure 1). Overall, MUN for AdMP was generally greater compared with the other treatments throughout the experiment, whereas differences between the other diets were inconsistent. However, a linear decrease in MUN ($P < 0.01$) was observed as MP supply decreased, but it was not affected by RP-His.

Dry matter, OM, and NDF intake during the sample collection for total-tract digestibility were not affected by the MP supplies (Table 4), but CP intake linearly decreased ($P < 0.01$) as the supply of MP decreased. Total-tract digestibility of CP linearly decreased ($P < 0.01$) as MP supply decreased. Total-tract digestibility of OM also tended to decrease linearly ($P = 0.09$) as MP supply decreased, whereas that of DM and NDF was not affected by the MP supply. Supplementation of RP-His did not affect intake and total-tract digestibility.

Reducing MP supply linearly decreased ($P \leq 0.01$) N intake, urinary N and urinary urea-N (UUN) excretion, and N balance (Table 5) but linearly increased ($P < 0.01$) fecal and milk N as a proportion of N intake. Reducing MP supply also tended to decrease ($P = 0.08$) UUN as a proportion of N intake. Uric acid excretion tended to be affected quadratically ($P = 0.09$) by MP supply. However, urinary excretion of allantoin and total purine derivatives was not affected by treatments. Supplementation of RP-His decreased ($P = 0.02$) MNE.

Blood creatinine and glucose were not affected by treatments (Table 6). Plasma albumin, alkaline phosphatase, aspartate aminotransferase, and nonesterified fatty acids were not affected by treatments. Blood urea nitrogen, plasma urea nitrogen (PUN), Hct, and Hb linearly decreased ($P \leq 0.01$) as MP supply decreased. Plasma globulin concentration decreased linearly ($P = 0.01$) and tended to decrease quadratically ($P = 0.07$) with decreasing MP supply. The ratio of albumin to globulin (AG ratio) increased linearly and quadratically ($P = 0.03$) with reducing MP supply. Total protein and GGT tended to decrease linearly ($P \leq 0.07$) as MP supply decreased. Plasma concentration of BHB was quadratically affected by MP supply ($P < 0.01$), in which BHB for MDMP was greatest. Hematocrit and Hb were not affected by the supplementation of RP-His. Supplementation of RP-His tended to increase ($P = 0.08$) plasma concentration of globulin but did not affect the AG ratio.

Plasma concentrations of Leu, Met, Val, and branched-chain AA (BCAA) decreased linearly ($P < 0.05$) as MP supply decreased (Table 7). Plasma His and Phe tended to linearly decrease ($P < 0.10$) with reducing MP supply. Plasma Pro was quadratically affected ($P = 0.02$), and Ser tended to be quadratically affected ($P = 0.09$) by MP supply, which was lowest for MDMP. Supplementation of RP-His decreased Met ($P < 0.01$) and tended to decrease Pro concentration ($P = 0.10$). Supplementation

Table 4. Effects of MP supply and rumen-protected His supplementation under adequate Met and Lys supplies on nutrient intake and total-tract digestibility of nutrients

Item	Diet ¹					SEM	P-value ²			
	AdMP	MDMP	MDMPH	DMP	DMPH		Overall	L	Q	His
Intake, kg/d										
DM	27.1	27.1	26.4	26.6	27.3	0.44	0.92	0.65	0.78	0.98
OM	25.4	25.4	24.7	25.0	25.6	0.41	0.92	0.69	0.82	1.00
NDF	7.57	7.78	7.45	7.97	8.24	0.145	0.16	0.24	0.97	0.89
CP ³	4.62 ^a	4.15 ^{bc}	4.17 ^b	3.85 ^c	3.99 ^{bc}	0.075	<0.01	<0.01	0.58	0.28
Total-tract digestibility, %										
DM	68.1	67.0	66.0	65.9	65.7	0.52	0.53	0.18	0.99	0.59
OM	68.7	67.1	66.6	66.0	66.6	0.49	0.48	0.09	0.86	0.95
NDF	40.2	38.8	37.1	36.6	40.4	0.91	0.58	0.22	0.87	0.62
CP	69.6 ^a	65.1 ^b	65.1 ^b	63.3 ^b	62.7 ^b	0.68	<0.01	<0.01	0.42	0.82

^{a-c}Means in a row with differing superscripts differ ($P < 0.05$).

¹AdMP, an MP-adequate diet that meets the MP, Met, Lys, and His requirements (NASEM, 2021); MDMP, an MP-moderately deficient diet (92.6% of the requirement) meeting the requirement of Met and Lys; MDMPH, the MDMP diet with RP-His to meet the His requirement; DMP, an MP-deficient diet (86.7% of requirement) meeting the requirement of Met and Lys; DMPH, the DMP diet with RP-His to meet the requirement of His.

²Overall, main effect of diets; L, linear effect of MP supply (AdMP vs. MDMP vs. DMP); Q, quadratic effect of MP supply (AdMP vs. MDMP vs. DMP); His, effect of His supplementation (MDMP, DMP vs. MDMPH, DMPH).

³N from RP-His was 5.91 g/d and 10.4 g/d for MDMPH and DMPH, respectively.

of RP-His did not affect plasma concentration of His, carnosine, and 1- or 3-methyl His.

Feces, urine, and manure excretion were not affected by treatments (Table 8). However, the ratio of feces to urine tended to decrease linearly ($P = 0.05$) as MP supply decreased. The daily CH₄ and CO₂ emission rates and their cumulative emissions were not affected by the MP supply. Supplementation of RP-His tended to decrease ($P \leq 0.10$) the daily emission rate and cumulative emission of CO₂. The rate of daily NH₃ emission showed an interaction ($P < 0.05$) between treatments and incubation days because the difference occurred mostly during the first 5 d of incubation. The daily NH₃ emission rate linearly decreased ($P \leq 0.01$) as MP supply decreased, but supplementation of RP-His did not affect the daily NH₃ emissions. Reducing MP supply also linearly decreased ($P < 0.01$) cumulative NH₃ emission.

DISCUSSION

Effects of MP Supply

We hypothesized that decreasing the MP supply below the requirement (92% to 86%) would negatively affect milk yield and milk protein yield, which did not occur in the current study. Production responses to reducing MP supply below the requirement are mixed in the literature. Milk yield and milk protein yield were not affected by reducing MP supply (90% to 96% of requirement according to NRC, 2001) without balancing AA in studies by Giallongo et al. (2015), Lee et al. (2015), and Räisänen et al. (2022). In contrast, decreases in milk yield and milk protein yield when cows were fed deficient MP diets (87% to

98% of the requirement according to NRC, 2001) without balancing AA in other studies (Lee et al., 2012c; Giallongo et al., 2016; Wang et al., 2022). In the current study, the lack of response in milk and milk protein yield might be explained by adequate supply of mMet and mLys. Previously, reducing MP supply but providing adequate mMet and mLys via RP-Met and Lys supplementation did not affect milk yield and milk protein yield in previous studies (Räisänen et al., 2022; Seleem et al., 2024). Under deficient MP supply, improving the profile of metabolizable AA by supplying adequate mMet and mLys may enhance the efficiency of absorbed AA utilization for production. In the current study, MNE increased from 31.4% to 38.3% as MP supply decreased, which occurred, at least in part, with decreased AA oxidation and excretion as supported by reduced BUN, MUN, and urinary N and UUN excretion as MP supply decreased. However, supplying adequate mMet or mMet and mLys under deficient MP did not always prevent reductions in milk yield and milk protein yield (Lee et al., 2012c; Giallongo et al., 2015, 2016). The inconsistent response to feeding adequate mMet and mLys under deficient MP might be, at least in part, explained by varying degrees of His deficiency among studies, which will be discussed later.

Another possible reason for the lack of the decrease in production despite reduced MP supply may have been that the MP deficiency was not as severe as that estimated by NASEM (2021). When both RDP and RUP decreased to lower dietary CP concentration and induced MP deficiency, decreases in total-tract NDF digestibility were often observed in studies (Lee et al., 2011, 2012c). In the current study, although NDF digestibility numerically decreased as MP supply decreased, it was not significant.

Table 5. Effects of MP supply and rumen-protected His supplementation under adequate Met and Lys supplies on N excretion and balance

Item ³	Diet ¹					SEM	P-value ²			
	AdMP	MDMP	MDMPH	DMP	DMPH		Overall	L	Q	His
N intake, g/d	739 ^a	663 ^{bc}	673 ^b	615 ^c	649 ^{bc}	11.9	<0.01	<0.01	0.58	0.28
Fecal N, g/d	227	234	234	226	244	6.3	0.86	0.96	0.63	0.47
Fecal N, % of N intake	30.4 ^b	34.9 ^a	34.9 ^a	36.7 ^a	37.3 ^a	0.68	<0.01	<0.01	0.42	0.82
Milk N, g/d	232	240	226	234	234	3.8	0.75	0.86	0.40	0.35
Milk N, % of N intake	31.4 ^c	36.4 ^{ab}	33.7 ^b	38.3 ^a	36.0 ^{ab}	0.57	<0.01	<0.01	0.24	0.02
Urine N, g/d	218 ^a	181 ^{abc}	192 ^{ab}	160 ^{bc}	150 ^c	7.5	0.01	0.01	0.64	0.98
Urine N, % of N intake	29.3 ^a	27.3 ^{ab}	28.1 ^a	26.0 ^{ab}	22.9 ^b	0.93	0.14	0.20	0.87	0.52
UUN, g/d	202 ^a	159 ^{bc}	175 ^{ab}	141 ^{bc}	130 ^c	7.1	<0.01	<0.01	0.47	0.86
UUN, % of urine N	92.6 ^a	87.9 ^{ab}	91.0 ^{ab}	88.0 ^{ab}	86.2 ^b	0.98	0.11	0.08	0.30	0.71
Uric acid, mmol/d	37.4	42.4	34.9	31.5	33.8	1.67	0.31	0.26	0.09	0.49
Allantoin, mmol/d	360	333	328	337	352	0.4	0.74	0.41	0.53	0.80
Total PD, mmol/d	397	375	363	368	386	12.0	0.77	0.33	0.76	0.91
N balance, g/d	63.3 ^a	8.62 ^b	21.0 ^b	-4.31 ^b	22.4 ^b	6.97	0.02	<0.01	0.24	0.18
N balance, % of N intake	8.92 ^a	1.31 ^b	3.31 ^b	-0.95 ^b	3.71 ^{ab}	1.13	0.05	<0.01	0.35	0.16

^{a-c}Means in a row with differing superscripts differ ($P < 0.05$).

¹AdMP, an MP-adequate diet that meets the MP, Met, Lys, and His requirements (NASEM, 2021); MDMP, an MP-moderately deficient diet (92.6% of the requirement) meeting the requirement of Met and Lys; MDMPH, the MDMP diet with RP-His to meet the His requirement; DMP, an MP-deficient diet (86.7% of requirement) meeting the requirement of Met and Lys; DMPH, the DMP diet with RP-His to meet the requirement of His.

²Overall, main effect of diets; L, linear effect of MP supply (AdMP vs. MDMP vs. DMP); Q, quadratic effect of MP supply (AdMP vs. MDMP vs. DMP); His, effect of His supplementation (MDMP, DMP vs. MDMPH, DMPH).

³Fecal excretion was estimated using iNDF; urine excretion was estimated using creatinine coefficient (23 mg of creatinine/kg of BW per day); UUN, urine urea-N; total PD, total purine derivatives; N balance = N intake - (Fecal N excretion + Urinary N excretion + Milk N excretion).

This is further supported by the lack of difference in urinary excretion of total purine derivatives, a commonly used proxy for MCP synthesis, despite some controversy over its reliability (Chen and Gomes, 1992; Reynal et al., 2005). In addition, although the N balance decreased with decreasing MP supply, the balance remained positive for cows on MDMP, and whereas it was negative for DMP, the deficit was minimal, that is, less than 1% of N intake. This aligns with the lack of differences in BW changes. If the MP deficiency was indeed overestimated by NASEM (2021), the factors contributing to this overestimation remain unclear. Previously, Lee et al. (2012b) reported that NRC (2001) underestimated MP-allowable milk for cows with a negative MP balance, with the degree of underestimation increasing as the MP balance decreased. It might be the case for NASEM (2021). To assess the potential underprediction of milk yield by NASEM (2021), we used a subset of the data from the regression analysis in Lee et al. (2012b), that is, individual cows fed MP-adequate ($n = 24$) or MP-deficient diets ($n = 24$) in Lee et al. (2011, 2012c), and performed a similar regression analysis using the NASEM (2021) model. As shown in Figure 2A, NASEM (2021) underpredicted the milk yield of cows fed MP-deficient diets (i.e., NASEM-predicted MP-allowable milk - actual milk yield), and the magnitude of underestimation increased as the negative MP balance became more severe ($R^2 = 0.99$). Compared with NRC (2001), the degree of underprediction by NASEM (2021) was nearly identical. Regression analysis of the individual-cow data from the current study (Figure 2B)

confirmed the same trend of underprediction ($R^2 = 0.99$) for cows fed MP-deficient diets.

Finally, no decrease in production despite deficient MP supply might be attributed to lack of His deficiency. We hypothesized that deficient MP supply would reduce milk and milk protein yields, even when mLys and mMet requirements are met, due to potential His deficiency (Lee et al., 2012c; Sok et al., 2017; Räisänen et al., 2023). As described earlier, His deficiency may occur in an MP-deficient dietary condition because of relatively low His content in MCP and reduced His supply from an MP-deficient diet. However, we did not observe any evidence supporting His deficiency for cows fed MDMP and DMP in the current study. Further details regarding the potential absence of His deficiency were discussed later in the context of the effects of RP-His supplementation.

Plasma Met concentration increased as MP supply decreased, which was not expected and inconsistent with previous studies (Lee et al., 2012c; Giallongo et al., 2015, 2016). In a study by Räisänen et al. (2022), RP-Met was added to all diets to ensure consistent mMet supply across treatments, with its dosage gradually increased as dietary MP supply decreased, as in the approach in the current study. In that study, although plasma Met concentration was not statistically different, plasma Met concentration was numerically greater (24.4 vs. 23.6 $\mu\text{mol/L}$, $P = 0.30$) for MP-deficient diets compared with MP-adequate diets. In the current study, a linear increase in plasma Met concentration was clear as MP supply decreased, which is difficult to explain. Increases in plasma Met

Table 6. Effects of MP supply and rumen-protected His supplementation under adequate Met and Lys supplies on blood metabolites

Item	Diet ¹					SEM	P-value ²			
	AdMP	MDMP	MDMPH	DMP	DMPH		Overall	L	Q	His
Blood metabolites ³										
BUN, mg/dL	12.2 ^a	10.5 ^{abc}	11.6 ^{ab}	8.6 ^c	9.2 ^{bc}	0.45	0.04	<0.01	0.94	0.37
CRE, µg/L	749	749	743	756	760	14.6	0.99	0.85	0.90	0.96
GLC, mg/dL	66.9	63.4	63.9	68.2	68.5	1.17	0.49	0.71	0.20	0.88
Hct, %PCV	25.4 ^a	24.5 ^{ab}	25.2 ^a	23.8 ^b	24.2 ^{ab}	0.22	0.05	<0.01	0.94	0.19
Hb, g/dL	8.64 ^a	8.34 ^{abc}	8.55 ^{ab}	8.08 ^c	8.25 ^{bc}	0.073	0.05	<0.01	0.92	0.19
Plasma metabolites ⁴										
ALP, U/L	67.6	68.7	64.1	63.2	65.3	2.62	0.56	0.25	0.32	0.65
AST, U/L	93.9	94.4	90.0	83.4	89.0	3.77	0.72	0.23	0.45	0.92
GGT, U/L	35.0	32.7	32.1	30.4	30.9	1.31	0.32	0.06	0.99	0.99
TP, g/dL	7.94	7.63	7.65	7.62	7.88	0.069	0.17	0.07	0.30	0.27
PUN, mg/dL	11.2 ^a	9.44 ^{abc}	10.7 ^{ab}	8.12 ^c	8.76 ^{bc}	0.349	0.01	<0.01	0.81	0.20
ALB, g/dL	4.05	4.11	4.05	4.07	4.06	0.026	0.92	0.75	0.46	0.51
GLOB, g/dL	3.94 ^a	3.55 ^b	3.75 ^{ab}	3.59 ^b	3.75 ^{ab}	0.061	0.04	0.01	0.07	0.08
AG ratio	1.04 ^c	1.17 ^a	1.08 ^{bc}	1.13 ^{ab}	1.13 ^{abc}	0.019	0.03	0.03	0.03	0.15
NEFA, µmol/L	97.4	97.0	106	97.0	102	4.14	0.93	0.97	0.99	0.43
BHB, µmol/L	825	1,240	924	745	752	65.7	0.05	0.65	0.01	0.23

^{a-c}Means in a row with differing superscripts differ ($P < 0.05$).

¹AdMP, an MP-adequate diet that meets the MP, Met, Lys, and His requirements (NASEM, 2021); MDMP, an MP-moderately deficient diet (92.6% of the requirement) meeting the requirement of Met and Lys; MDMPH, the MDMP diet with RP-His to meet the His requirement; DMP, an MP-deficient diet (86.7% of requirement) meeting the requirement of Met and Lys; DMPH, the DMP diet with RP-His to meet the requirement of His.

²Overall, main effect of diets; L, linear effect of MP supply (AdMP vs. MDMP vs. DMP); Q, quadratic effect of MP supply (AdMP vs. MDMP vs. DMP); His, effect of His supplementation (MDMP, DMP vs. MDMPH, DMPH).

³CRE, creatinine; GLC, glucose; Hct, hematocrit (% of PCV, packed cell volume); Hb, hemoglobin (g/dL) = Hct (% of PCV) $\times$ 0.34.

⁴ALP, alkaline phosphatase; AST, aspartate aminotransferase; GGT, gamma-glutamyl transferase; TP, total protein; PUN, plasma urea nitrogen; ALB, albumin; GLOB, globulin; AG ratio, albumin-to-globulin ratio; NEFA, nonesterified fatty acids; BHBA, BHB acid.

concentration can occur due to either increased supply or decreased utilization of Met in circulation. Perhaps, the utilization of Met in circulation might have decreased in tissues under MP deficiency if other AA were deficient. Indeed, Leu, Phe, and Val concentrations decreased as MP decreased. In addition to that, the inclusion of RP-Met increased in the diets as MP decreased, leading to a greater Met supply relative to other AA. The reason for increased plasma Met concentration as MP decreased cannot be determined in the current study, and further studies are needed to confirm the results.

We observed a linear decrease in total-tract CP digestibility when MP decreased, which likely occurred due to a dilution of metabolic fecal N with undigested feed N excretion (Broderick, 2003; Colmenero and Broderick, 2006). Decreasing MP supply tended to decrease total-tract OM digestibility, which likely occurred due to numerical decreases in total-tract NDF digestibility as MP decreased. Decreases in total-tract NDF digestibility have often been observed in studies when dietary protein or MP supply decreased because of limited RDP supply (Lee et al., 2012c; Giallongo et al., 2015; Räisänen et al., 2022).

Effects of RP-His Supplementation

Supplementation of RP-His to the MP-deficient diets affected neither DMI nor any production measures in the

current study. The lack of His effect perhaps may have occurred because His was not deficient. In the current study, we did not find any clear evidence supporting His deficiency for cows fed the MP-deficient diets. For example, plasma His concentration decreased as MP supply decreased, which may have occurred due to decreased supplies of mHis from dietary or microbial His supply. However, plasma concentrations of His for AdMP, MDMP, and DMP were 57.6, 52.8, and 48 µmol/L, respectively, and those are at the high end in the range of plasma His concentration reported in His studies (Räisänen et al., 2023), reflecting that His was sufficiently available in blood circulation. If that was the case, an increase in plasma His concentration from the additional supply of mHis may have been minimal because plasma His concentration responds curvilinearly to mHis supply due to increasing liver oxidation (Lapierre et al., 2021), suggesting that an increase in plasma His concentration would be minimal when mHis supply is not deficient. Indeed, no change in plasma His concentration was observed when cows were fed an MP-adequate diet supplemented with varying levels of RP-His (Räisänen et al., 2021b), where DMI and production were not affected by RP-His supplementation. Plasma His concentration is a result of supply, oxidation, and tissue uptake. Therefore, the lack of increase in plasma His concentration with RP-His may have been due to increases in tissue uptake

Table 7. Effects of MP supply and rumen-protected His supplementation under adequate Met and Lys supplies on plasma AA concentrations (μmol/L)

Item ³	Diet ¹					SEM	<i>P</i> -value ²			
	AdMP	MDMP	MDMPH	DMP	DMPH		Overall	L	Q	His
Arg	95.8	96.0	101	95.8	90.9	3.34	0.93	0.99	0.98	0.99
His	57.6	52.8	54.9	48.0	49.4	1.74	0.31	0.06	1.00	0.64
Ile	158	153	153	151	143	3.5	0.78	0.55	0.85	0.67
Leu	201 ^a	168 ^{bc}	181 ^{ab}	163 ^{bc}	139 ^c	6.0	0.01	0.02	0.32	0.64
Lys	104	102	106	106	94.7	3.02	0.75	0.82	0.70	0.59
Met	37.7 ^b	44.8 ^b	39.5 ^b	56.0 ^a	44.3 ^b	1.56	<0.01	<0.01	0.59	<0.01
Phe	56.6 ^{ab}	50.7 ^{bc}	58.9 ^a	50.2 ^{bc}	47.2 ^c	1.41	0.02	0.10	0.43	0.36
Thr	108	112	118	107	106	3.0	0.67	0.95	0.63	0.72
Trp	47.0	45.7	49.3	47.4	45.3	1.4	0.85	0.93	0.66	0.80
Val	353 ^a	306 ^{abc}	322 ^{ab}	297 ^{bc}	265 ^c	8.5	0.01	0.02	0.39	0.64
BCAA	711 ^a	626 ^{abc}	655 ^{ab}	610 ^{bc}	547 ^c	17.4	0.03	0.05	0.45	0.65
EAA	1,216	1,130	1,183	1,117	1,031	28	0.26	0.24	0.62	0.79
Ala	300 ^{ab}	270 ^b	296 ^{ab}	334 ^a	314 ^{ab}	9.4	0.17	0.17	0.04	0.88
Asn	42.3	43.1	46.7	45.4	41.2	1.10	0.49	0.37	0.81	0.89
Asp	10.2	10.4	9.46	9.58	8.45	0.512	0.57	0.63	0.62	0.25
Gln	250	241	259	274	241	8.2	0.70	0.37	0.35	0.69
Glu	65.9	63.8	57.5	61.2	57.5	2.16	0.47	0.41	0.96	0.22
Gly	324	342	336	347	331	11.4	0.92	0.39	0.80	0.57
Pro	110 ^a	89.9 ^b	93.1 ^{ab}	111 ^a	83.3 ^b	3.52	0.03	0.96	0.02	0.10
Ser	109	97.2	111	110	103	2.5	0.39	0.97	0.09	0.50
Tau	58.6	57.5	57.8	64.9	59.1	2.28	0.84	0.37	0.50	0.59
Tyr	59.7	54.1	60.7	52.4	48.7	2.30	0.43	0.31	0.76	0.78
NEAA	1,267	1,198	1,266	1,330	1,263	26.3	0.64	0.44	0.17	0.99
Total AA	2,480	2,314	2,447	2,445	2,304	48.3	0.70	0.82	0.28	0.97
Car	13.6 ^{bc}	12.8 ^c	13.1 ^{bc}	15.9 ^{ab}	16.6 ^a	0.55	0.03	0.12	0.13	0.62
1-M-His	3.17	3.37	3.56	3.60	4.18	0.147	0.22	0.31	0.98	0.22
3-M-His	3.84	3.22	3.00	3.69	3.07	0.136	0.14	0.72	0.12	0.15

^{a-c}Means in a row with differing superscripts differ ($P < 0.05$).

¹AdMP, an MP-adequate diet that meets the MP, Met, Lys, and His requirements (NASEM, 2021); MDMP, an MP-moderately deficient diet (92.6% of the requirement) meeting the requirement of Met and Lys; MDMPH, the MDMP diet with RP-His to meet the His requirement; DMP, an MP-deficient diet (86.7% of requirement) meeting the requirement of Met and Lys; DMPH, the DMP diet with RP-His to meet the requirement of His.

²Overall, main effect of diets; L, linear effect of MP supply (AdMP vs. MDMP vs. DMP); Q, quadratic effect of MP supply (AdMP vs. MDMP vs. DMP); His, effect of His supplementation (MDMP, DMP vs. MDMPH, DMPH).

³BCAA, branched-chain AA; Tau, taurine; Car, carnosine; 1-M-His, 1-methylhistidine; 3-M-His, 3-methylhistidine.

of His. In previous studies, muscle His concentration decreased under His deficiency (Giallongo et al., 2017) and increased with additional supplies of mHis (Zang et al., 2019; Räisänen et al., 2022). However, we did not measure muscle His concentration in the current study, and BW and BW change were not affected by RP-His in the current study.

We also did not observe any changes in blood hemoglobin and plasma 1-methyl His and carnosine by RP-His supplementation. In contrast, Räisänen et al. (2021a) reported that varying levels of RP-His supply with an MP-deficient diet (87% of the MP requirement and 37–74 g/d of mHis according to NRC, 2001) linearly increased milk yield, milk protein yield, and plasma His, 1-methyl His, and carnosine concentration. In that study, plasma His concentration for the MP-deficient diet without RP-His supplementation was 21 μmol/L, which is similar to the level observed in another study (26 μmol/L) on feeding an MP-deficient diet (87% of the MP requirement and 44 g/d of mHis according to NRC, 2001) without RP-His (Lee et al., 2012c). Plasma His concentrations for MP-deficient

diets in the studies above were much lower than those observed for the MP-deficient diets in the current experiment (92% and 86% of the MP requirement, and 66 and 58 g/d of mHis for MDMP and DMP, respectively, according to NASEM, 2021). It is difficult to explain the lack of deficiency of His for the MP-deficient diets if that was the case. In the current experiment, however, blood meal was used as one of the protein sources for all diets and is known for high RUP content and high His concentration. Therefore, the supplies of mHis for MDMP (15.4% CP) and DMP (14.5% CP) were 66 and 58 g/d, respectively, and those were relatively greater compared with those for diets with similar CP concentration in previous studies, e.g., 54 g/d for a 15.7%-CP diet and 44 g/d for a 13.6%-CP diet in Lee et al. (2012c), and 64 g/d for a 16.6%-CP diet and 57 g/d for a 15.3%-CP diet in Räisänen et al. (2022) based on NRC (2001). The inclusion of blood meal was appropriate in the current study to evaluate the effect of RP-His based on NASEM (2021). However, previous studies assumed the mHis requirement to be 2.2% of MP, and the required mHis at the target His utilization

Table 8. Effects of MP supply and rumen-protected His supplementation under adequate Met and Lys supplies on manure and accumulated GHG emissions from manure

Item	Diet ¹					SEM	P-value ²			
	AdMP	MDMP	MDMPH	DMP	DMPH		Overall	L	Q	His
Feces, kg/d	61.8	67.6	59.3	66.1	68.2	2.30	0.31	0.39	0.39	0.38
Urine, kg/d	27.6	25.6	23.7	24.0	24.2	0.95	0.63	0.21	0.94	0.69
Manure, kg/d	89.4	93.2	83.1	90.0	92.4	2.89	0.56	0.93	0.54	0.40
Feces:urine	2.29	2.73	2.61	2.86	2.79	0.096	0.30	0.05	0.55	0.62
CH ₄ , g/d per cow	0.648	0.667	0.537	0.560	0.622	0.0294	0.35	0.24	0.34	0.52
CO ₂ , g/d per cow	198	204	168	180	175	3.2	0.18	0.29	0.32	0.10
NH ₃ , g/d per cow	11.6 ^a	11.1 ^a	9.18 ^a	7.51 ^b	7.52 ^b	0.320	0.02	0.01	0.24	0.37
Cumulative CH ₄ , g	6.46	6.72	5.45	5.46	6.21	0.464	0.32	0.19	0.25	0.62
Cumulative CO ₂ , kg	2.38	2.43	2.00	2.15	2.08	0.774	0.10	0.21	0.29	0.06
Cumulative NH ₃ , g	122 ^a	118 ^a	97.6 ^{ab}	80.7 ^b	81.8 ^b	5.65	0.02	<0.01	0.19	0.36

^{a,b}Means in a row with differing superscripts differ ($P < 0.05$).

¹AdMP, an MP-adequate diet that meets the MP, Met, Lys, and His requirements (NASEM, 2021); MDMP, an MP-moderately deficient diet (92.6% of the requirement) meeting the requirement of Met and Lys; MDMPH, the MDMP diet with RP-His to meet the His requirement; DMP, an MP-deficient diet (86.7% of requirement) meeting the requirement of Met and Lys; DMPH, the DMP diet with RP-His to meet the requirement of His.

²Overall, main effect of diets; L, linear effect of MP supply (AdMP vs. MDMP vs. DMP); Q, quadratic effect of MP supply (AdMP vs. MDMP vs. DMP); His, effect of His supplementation (MDMP, DMP vs. MDMPH, DMPH).

³Interaction between diets and day was significant ($P = 0.02$).

efficiency recommended by NASEM (2021) was about 2.6% of MP. If the true mHis requirement was close to 2.2% of MP in the current study, cows fed the MDMP and DMP diets would have needed 58 and 53 g/d as the mHis requirement, respectively, and would have provided sufficient mHis without RP-His supplementation, explaining the lack of His deficiency in the current study.

The lack of response to RP-His may have occurred due to the length of the experiment. Histidine can be provided from endogenous His sources such as muscle carnosine and blood Hb, thus, feeding a His-deficient diet to cows for the short-term may not develop His deficiency (Lapierre et al., 2021). Giallongo et al. (2017) discussed that muscle carnosine and blood Hb can theoretically provide sufficient His for 7 wk under feeding a His-deficient diet. When experiments were conducted in a randomized complete block design for 7 wk or more (Lee et al., 2012c; Giallongo et al., 2016, 2017), DMI, milk yield, and milk protein yield increased, and plasma His concentrations were elevated when RP-His was provided with a His-deficient or MP-deficient diet. The current experiment was conducted for 6 wk, and His deficiency for the MP-deficient diets may not have occurred due to His supplies from the endogenous sources. In short-term studies (a Latin square design with 28 d per period), however, positive responses of milk and milk protein yield were observed when RP-His was provided under deficient MP supply (Zang et al., 2019; Räisänen et al., 2021a) or adequate MP supply (Morris and Kononoff, 2020). Furthermore, Räisänen et al. (2022) examined the effects of RP-His supplementation with an MP-adequate and deficient diet for 10 wk. Whereas RP-His tended to increase ($P = 0.10$) milk

yield regardless of MP levels, DMI and milk component yields were not affected by RP-His. In that study, the MP-deficient diet was deficient in MP (97% of the requirement; NRC, 2001), less than expected (85% of the requirement; NRC, 2001). According to results in the literature, the experimental length and the degree of MP deficiency do not fully explain the lack of RP-His effect observed in the current study. Perhaps His deficiency rather than MP deficiency was more associated with responses to feeding RP-His. In a study by Giallongo et al. (2017), cows were fed either an MP- and His-adequate diet or an MP-adequate but His-deficient diet for 8 wk. The concentrations of His in the diets were adjusted by varying the proportion of porcine blood meal (His-rich) or feather meal (His-deficient). Although MP was adequate for both treatment groups, cows on the His-adequate diet increased DMI, milk yield, and milk protein yield compared with those on the His-deficient diet. In that study, plasma His concentration was 37.3 $\mu\text{mol/L}$ for cows on the His-deficient diet, which was lower than the concentration of His observed for cows on DMP in the current study (48.0 $\mu\text{mol/L}$).

Endogenous His supplies may have resulted in the lack of His deficiency for MDMP and DMP. Hemoglobin and carnosine have been mainly considered endogenous sources of His. In the current study, lowering MP supply linearly decreased blood Hb from 8.64 (AdMP) to 8.08 g/dL (DMP), which is in line with Giallongo et al. (2016). However, it is uncertain whether Hb actually served as an endogenous His source. For example, blood Hb decreased for DMP from 8.51 g/dL in the co-variate period to 8.04 g/dL in wk 6 of the experimental period. If the decreased Hb was used to provide His,

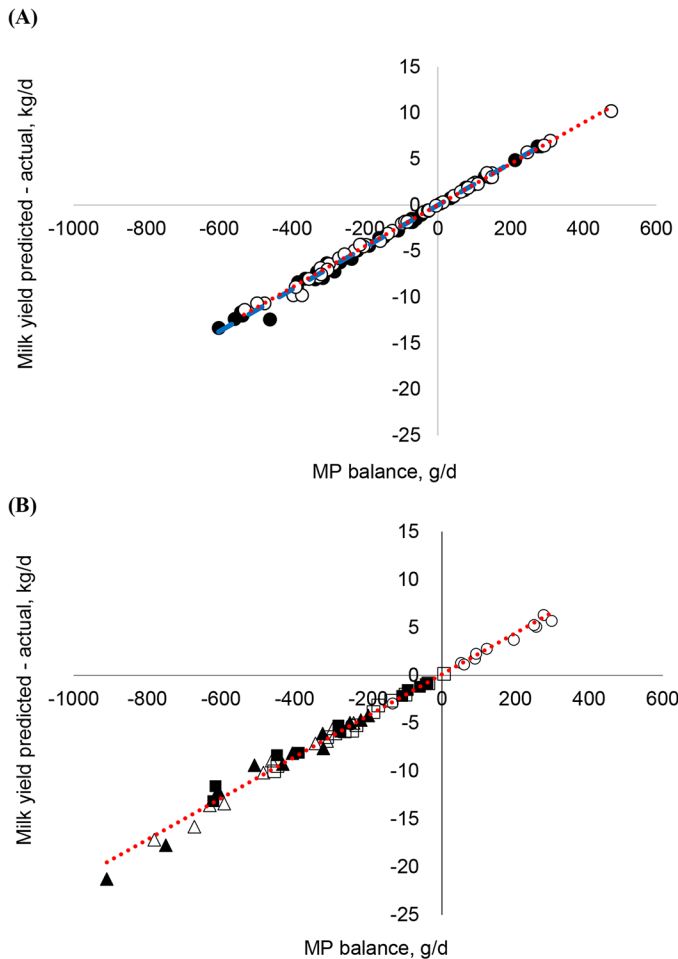


Figure 2. (A) A linear relationship between MP balance and the difference in milk yield between predicted and actual from previous studies of Lee et al. (2011) and Lee et al. (2012c). Milk yield predicted was obtained from NASEM-MP-allowable milk yield (closed circle) or NRC-MP-allowable milk yield (opened circle). NASEM (2021): $y = 0.0228 \times \text{MP balance} - 0.0182$ ($R^2 = 0.99$). NRC (2001): $y = 0.0224 \times \text{MP balance} + 0.0024$ ($R^2 = 0.99$). Data points represent individual cows fed either an MP-deficient diet ($n = 24$) or MP-adequate diet ($n = 24$) in Lee et al. (2011) and Lee et al. (2012c). (B) A linear relationship between MP balance and the difference in milk yield between predicted and actual from the current study. Milk yield predicted was obtained from NASEM-MP-allowable milk yield based on actual diet composition, DMI, and BW. $y = 0.0216 \times \text{MP balance} + 0.1048$ ($R^2 = 0.99$). The opened circle indicates an MP-adequate diet ($n = 12$), closed square indicates an MP-moderately deficient diet ($n = 11$), opened square indicates an MP-moderately deficient diet supplemented with rumen-protected His ($n = 12$), closed triangle indicates an MP-deficient diet ($n = 11$), opened triangle indicates an MP-deficient diet supplemented with rumen-protected His ($n = 11$).

theoretically, it might have supplied 170 mg of His (according to the calculation in Räsänen et al., 2021b), which is unlikely to be sufficient for additional His to meet the requirement for the DMP. In addition, studies are available where incremental His supply under various MP supplies did not affect blood Hb concentra-

tion (Lapierre et al., 2021; Räsänen et al., 2021a,b). Likewise, the response of plasma carnosine to mHis supply also varies in the literature. Plasma carnosine concentration increased with RP-His or abomasal infusion of His in some studies (Zang et al., 2019; Lapierre et al., 2021; Räsänen et al., 2021a), but not in others (Lee et al., 2012c; Giallongo et al., 2016; Räsänen et al., 2022). Further studies are needed to determine the contribution of endogenous His sources to His supply for milk protein synthesis under varying His deficiencies. We observed that plasma globulin concentration decreased as MP supply decreased and that tended to increase by RP-His supplementation. Various globulin proteins that are involved in transport and enzymatic functions contain His residues (Busher, 1990). Perhaps certain globulin proteins may have served as a source of endogenous His. The potential role of globulin proteins as an endogenous His source warrants further studies.

One may question the accuracy of His deficiency for the MP-deficient diets that were estimated by NASEM (2021) in the current experiment. Recently, a meta-analysis by Martineau et al. (2024) compared the prediction of postruminal outflows of EAA and their prediction errors among requirement models (NRC, 2001; NASEM, 2021; CNCPS, v6.5.5, Van Amburgh et al., 2015). In that study, the mean value of postruminal His outflow was lower (56 vs. 65 g/d) when predicted by NASEM (2021) compared with CNCPS. In addition, when His balances for MDMP and DMP in the current experiment were predicted by CNCPS, those diets were not deficient in His (data not shown). As discussed in Martineau et al. (2024), the greater postruminal His outflow in CNCPS compared with NASEM is primarily due to the higher His concentration used in MCP in CNCPS. If the prediction of His supply by CNCPS is more accurate than that of NASEM (2021), no response to RP-His in the current study is likely plausible, as positive production responses to additional mHis are more likely to occur under His-deficient conditions as discussed previously. However, in the meta-analysis by Martineau et al. (2024), it appears that the prediction errors were greater for CNCPS versus NASEM, and the authors concluded that CNCPS overestimates the postruminal outflow of His.

Dietary N use efficiency for milk N (MNE) decreased by RP-His because RP-His did not increase milk protein yield, whereas it increased N intake. Most previous RP-His studies reported that RP-His did not affect MNE (Lee et al., 2012c; Giallongo et al., 2016; Räsänen et al., 2021a). Studies in which RP-His did not affect milk protein yield still reported unchanged MNE by RP-His (Räsänen et al., 2021b, 2022). However, in those studies, N intake did not increase with RP-His supplementa-

tion even if a greater amount of RP-His was included in the diets than in the current study.

Gas Emissions from Manure

The increasing tendency for the feces-to-urine ratio (kg/kg, as-is basis) with decreasing MP supply was expected. Dietary N supply is linked to urine output, as urination increases to excrete excess N or N that was not used for anabolic purposes from the body (Weiss et al., 2009; Dijkstra et al., 2013). The linear decrease in daily NH_3 emission rate and cumulative NH_3 emissions with decreasing MP supply are consistent with previous studies. Külling et al. (2001) reported that lowering CP from 17.5% to 12.5% decreased NH_3 emitting potential from manure by 74%. Lee et al. (2012a) observed a decrease in NH_3 emission from dairy manure by 47% when dietary CP decreased from 16.7% to 14.8% (DM basis). Another study by Lee et al. (2012b) found a 37% reduction in NH_3 emission from manure when dietary CP decreased from 15.6 to 14.0% (DM basis). In the current study, dietary CP decreased from 17.0 (AdMP) to 14.6% (DMP), and the cumulative NH_3 emission decreased by 33%. Despite the methodological differences in measuring NH_3 emissions from manure, e.g., soil application, incubation length, feces-to-urine ratio, and the amount of manure used for incubation, consistent findings indicate that lowering dietary CP is an effective strategy for reducing NH_3 emissions from manure. The supplementation of RP-His did not increase NH_3 emissions, which is in line with the results in Lee et al. (2012b), where NH_3 emissions from manure of cows fed a diet supplemented with or without RP-Met did not differ. Interestingly, RP-His supplementation tended to decrease the daily CO_2 emission rate and the cumulative CO_2 emission from manure, which is difficult to explain. Carbon dioxide is produced from manure during the process of microbial decomposition. No change in CH_4 emission with decreased CO_2 likely indicates that aerobic decomposition was affected more than anaerobic decomposition by RP-His supplementation during the manure incubation. The RP-His used in the current study was a fat-coated His, and therefore RP-His supplementation slightly increased fat supply. However, the additional supply of fat from RP-His to the diets was trivial (about <0.1% of dietary DM), and even though the fat from RP-His was assumed to be 100% excreted in manure, it did not seem to be sufficient to affect the bacterial decomposition of manure OM. Further studies are needed to confirm our results.

CONCLUSIONS

Lowering MP supply increased MNE and decreased urinary N excretion and NH_3 emissions from manure,

but it did not affect milk yield and composition and BW change. The lack of a decrease in milk yield and milk protein yield with the deficient MP supplies likely occurred due to adequate mLys and mMet supplies without His deficiency. Although reducing MP supply decreased plasma His and blood Hb concentration, RP-His supplementation did not affect plasma His and carnosine concentration and blood Hb concentration, suggesting that His supply for milk protein synthesis was likely not deficient during the experiment. The absence of His deficiency with reduced MP supplies may have occurred due to the relatively short experiment period (6 wk) during which additional His may have been provided from endogenous His sources. As a result, RP-His supplementation to the MP-deficient diets with RP-His did not improve intake and milk production. Further studies to identify endogenous His sources and determine their contribution to mHis for milk protein synthesis are needed.

NOTES

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Nonstandard abbreviations used: AdMP = MP-adequate diet supplemented with RP-Met that meets the NASEM requirements for MP, mMet, mLys, and mHis; AG = albumin-to-globulin ratio; BCAA = branched-chain AA; CRE = creatinine; DMP = MP-deficient diet supplemented with RP-Met and Lys that meets 85% of MP, 105% of Met and Lys, and 81% of His required; DMPH = MDP diet with 79 g/d of RP-His to meet the requirement of His; GLC = glucose; Hb = hemoglobin; Hct = hematocrit; iNDF = indigestible NDF; MCP = microbial protein; MDMP = an MP-moderately deficient diet supplemented with RP-Met and Lys that meets 93% of MP, 105% of mMet and mLys, and 91% of mHis required; MDMPH = 45 g/d of RP-His to meet the His requirement; mHis = metabolizable His; mLys = metabolizable Lys; mMet = metabolizable Met; MNE = increased N use efficiency

for milk protein; NEFA = nonesterified fatty acids; PCV = packed cell volume; PUN = plasma urea nitrogen; RP = rumen-protected; total PD, total purine derivatives; UUN = urinary N and urinary urea-N.

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